

Theory of Computation

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Introduction

What is Formal Language ?

- It is the mathematical study of computing machine and its capabilities.
- It is a study of formal language and automata theory.

Alphabet

- The non empty finite set of symbols is called as an alphabet and it is denoted by Σ .
 - Example
 - $\Sigma = \{a.b.c.....z\}$
 - $\Sigma = \{0,1\}$
 - $\Sigma = \{1\}$

String

- Any finite sequence of symbol from alphabet Σ , is called as string and is represented by w .
 - Example
 - $\Sigma = \{a,b\}$
 - $W = a,ab,aa,bb$
 - $W = ab,aab,abc$
 - $\Sigma = \{0,1\}$
 - $W = 0,01,00,11,1$
 - $W = 102,2013$
 - $\Sigma = \{1\}$
 - $W = 1, 11, 111$
 - $W = 1,12, 121$

Length of a String

- If w is any string over alphabet Σ then the number of symbols involved in the sequence of string is called as length of the string and denoted by $|w|$.
- **Example**
 - $\Sigma = \{a,b\}$ $W = a, ab, aa, bb$, $|w| = 1, 2, 2, 2$
 - $\Sigma = \{0,1\}$ $W = 0, 01, 00, 11, 1$ $|w| = 1, 2, 2, 2$
- **Empty String**
- A string of length zero or string without any symbols is known as empty string and is denoted by ϵ
 - $w = \epsilon$, $|w| = 0$
 - $w. \epsilon = w = \epsilon. w$

Substring

- Let u, w be the two strings over same alphabet Σ then u is said to be substring of w if u can be obtained from w .
- Any consecutive sequence of symbols over given string.
 - If u is a substring of w then $|u| \leq |w|$
 - Every string is substring to itself.
 - Empty string “ ϵ ” is substring for every string.
- **Example**
 - $W = \text{TOC}$
 - Zero length Substring = ϵ One length substring = T, O, C
 - Two length substring = TO, OC Three length substring = TOC

Types of Substring

- **The substrings are classified into two types**
 1. Trivial Substring or improper Substring
 2. Non-trivial or proper substring
- **Trivial Substring or improper Substring**
 - If w is any string over alphabet Σ then the substring w itself and ϵ is called as trivial substring
- **Non-trivial Substring or proper Substring**
 - If w is any string over alphabet Σ then any substring of w other than w itself and ϵ is called non trivial substring.

Facts about Substring

- **If w is any string with distinct symbols and $|w| = n$**
 1. No of substrings = $\sum_{i=1}^n i = n(n+1)/2$
 2. No. of trivial string = 2
 3. No. of non trivial substring = $\sum_{i=1}^n i - 1$
 4. No. of non empty substring = $\sum_{i=1}^n i$
 5. No of substrings of distinct length = n
 6. No. of strings of length n generated over alphabet $\Sigma = |\Sigma|^n$

Prefix and Suffix

- **Prefix**
 - The sequence of starting or leading symbol is called as prefix.
- **Suffix**
 - The sequence of ending or trailing symbol is called as suffix.
- **Example**
- $w = \text{TOC}$, $|w| = 3$
- Prefix : ϵ , T , TO , TOC
- Suffix : TOC , OC , C , ϵ

Facts about Prefix and Suffix

- If w is any string of length ' n ' then
 1. No. of prefix = No. of suffix = $n+1$
 2. Trivial substrings are common for both prefix and suffix
 3. Every prefix and suffix must be a substring but every substring need not be prefix or suffix.
- Prefixes and suffixes play crucial roles in various string processing algorithms and are fundamental concepts in computer science, particularly in fields such as pattern matching, text processing, and data compression.

Power of an alphabet

- If Σ is any alphabet then Σ^k is the set of all the strings of length k .
- Example : $\Sigma = \{0,1\}$
 - $\Sigma^2 = \{00,01,10,11\}$
 - $\Sigma^3 = \{000,001,010,011,100,101,110,111\}$
 - $\Sigma^k = \{\text{all } k\text{-length strings}\}$
- **+ve closure(Σ^+)**
 - $\Sigma^+ = \{w \mid |w| \geq 1\}$
- **Kleen closure(Σ^*)**
 - $\Sigma^* = \{w \mid |w| \geq 0\}$
 - $\Sigma^* = \Sigma^+ \cup \epsilon$

Language

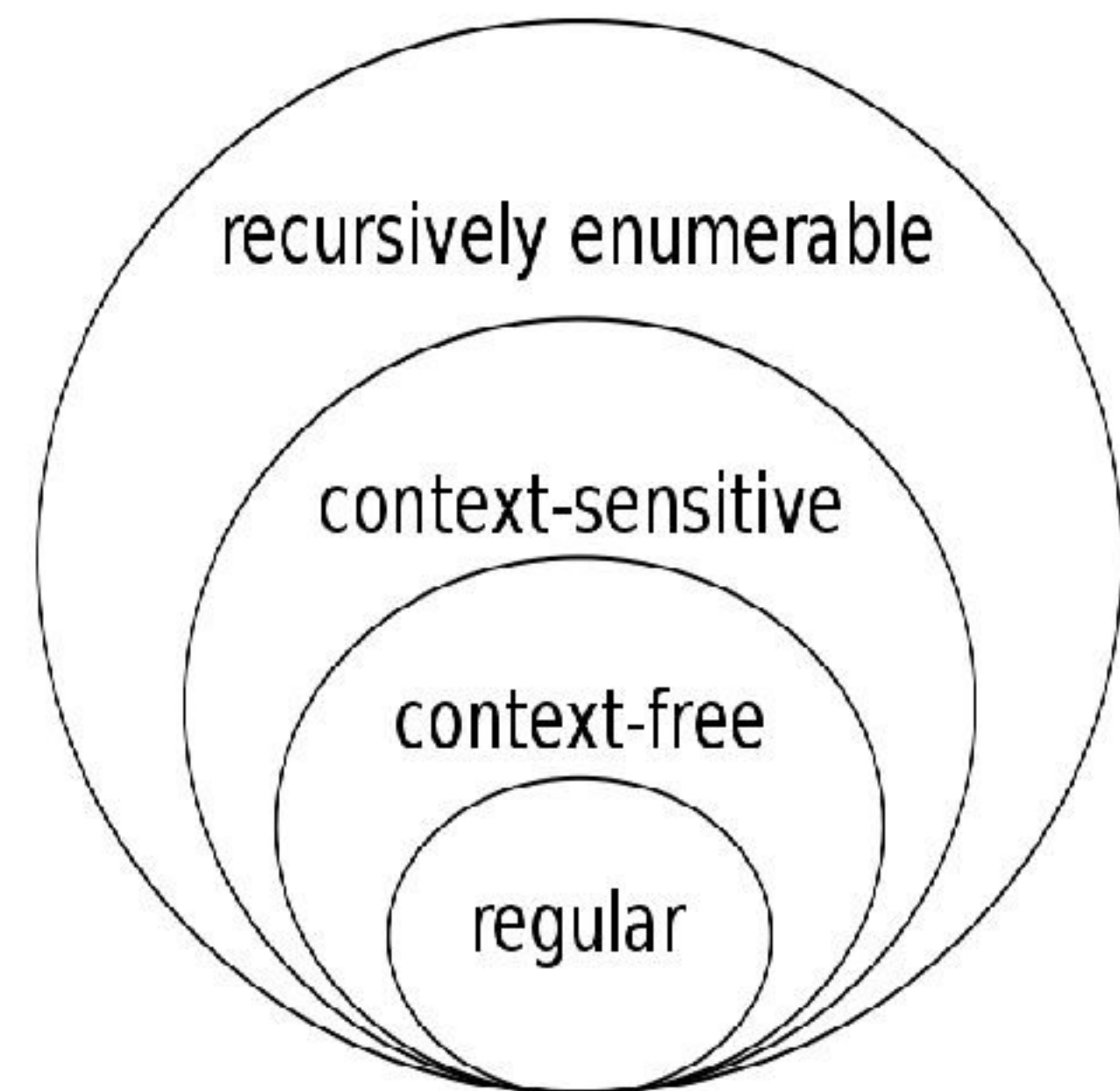
- The collection of strings from the alphabet Σ is called language
 - Example $\Sigma = \{0,1\}$
 - $L = \{00,01,10,11\}$
 - $L = \{ (01)^n \mid n \geq 1 \}$
 - $L = \{ 0^n 1^m \mid m \geq 1, n \geq 1 \}$
- If Σ is any alphabet then Σ^* is called as universal language

Formal Language

- The collection of strings where we can put some restriction in the formation of string is called as formal language.
 - Example $\Sigma = \{0,1\}$
 - $L = \{00,01,10,11\}$
 - $L = \{ (01)^n \mid n \geq 1 \}$
 - $L = \{ 0^n 1^m \mid m \geq 1, n \geq 1 \}$

Chomsky classification of Formal Language

- According to Chomsky the formal languages are classified as
 1. Type 0 or Recursive Enumerable Languages
 2. Type 1 or Context sensitive languages
 3. Type 2 or Context free languages
 4. Type 3 or Context Regular languages



Types of Language

- **Empty Languages**
 - The Language that does not contain any string even empty string is called as empty language and is denoted by ϕ .
- **Non Empty Languages**
 - The Language that contain at least one string is called as non empty language .
- **Finite Languages**
 - The Language which contains finite number of strings and length of each string is finite is called as finite language .
 - $L = \{ 0^n 1^n \mid 1 \leq n \leq 1 \}$
- **Infinite Languages**
 - The Language which contains infinite number of strings and length of each string is finite is called as infinite language .
 - $L = \{ 0^n 1^n \mid n \geq 1 \}$

Automata

- The mathematical system that can represent the formal language is called as Automata i.e. The mathematical representation of formal language is called as an automata.
- **Types of Automata**
 1. Finite Automata(FA)
 2. Push Down Automata(PDA)
 3. Linear Bound Automata(LBA)
 4. Turing Machine(LBA)

Expressive Power of an Automata

- The number of languages accepted by an automata is called as Expressive Power of an Automata.
 1. $E(FA)=1$
 2. $E(PDA)=2$
 3. $E(LBA)=3$
 4. $E(TM)=4$

Grammar

- The collection of rules which are used to generate the string is called grammar.
- Grammar G is a collection of 4 tuples {V,T,P,S}

G= {V,T,P,S} ,Where

V= set of all Nonterminal symbol/variable

P= set of all productions

T= set of all terminal symbols

S= starting symbol

■ Example

1. $A \longrightarrow XYZ$ (r1) $V=\{A,X,Y,Z\}$
 $X \longrightarrow a$ (r2) $T=\{a,b,c\}$
 $Y \longrightarrow b$ (r3) $P=\{r1,r2,r3,r4\}$
 $Z \longrightarrow c$ (r4) $S=A$

Chomsky classification of Grammar

- According to Chomsky the Grammar is classified as

1. Type 0 or Recursive Enumerable Grammar
2. Type 1 or Context sensitive Grammar
3. Type 2 or Context free Grammar
4. Type 3 or Context Regular Grammar

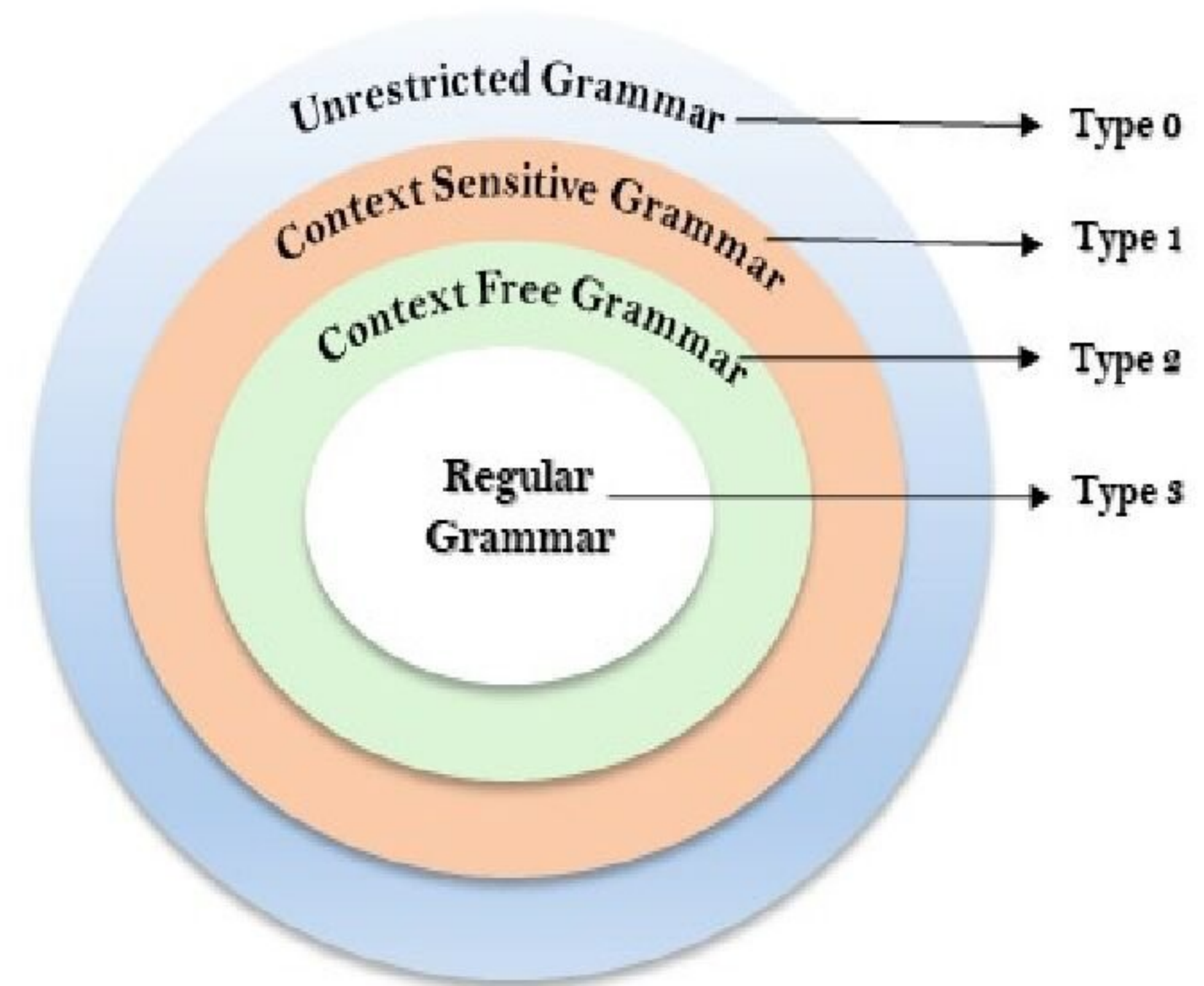


Fig: Chomsky Hierarchy

Other classification of Grammar

- **Recursive grammar**
 - The grammar g is said to be recursive if atleast one production contain the same variable at both left hand side and right hand side of production.
 - **Example** $S \longrightarrow asb | \epsilon$
- **Non Recursive grammar**
 - The grammar g is said to be recursive if no production contain the same variable at both left hand side and right hand side of production.
 - **Example** $S \quad ab | \epsilon$

Derivation

- The process of deriving a string is called as derivation.
- The geometrical representation of derivation is called Derivation tree or parse tree.
- Steps involved in derivation is called sentential form

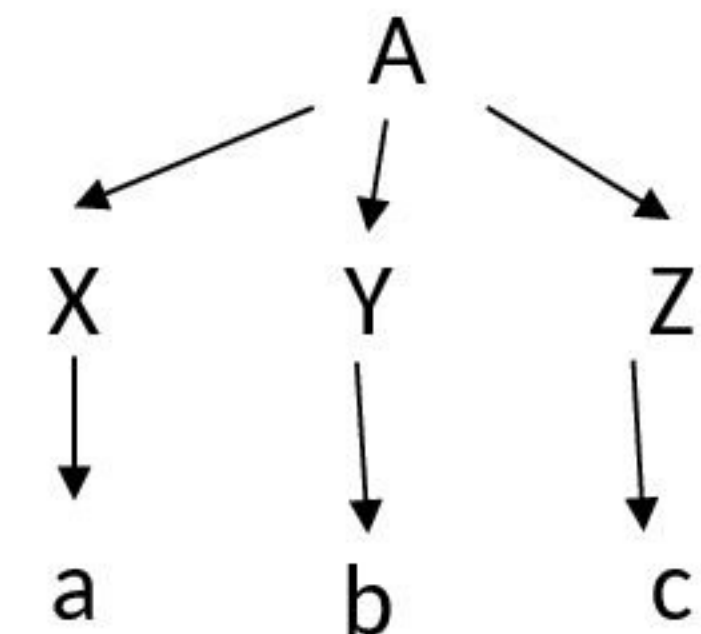
• Example

1. $A \longrightarrow XYZ$
 $X \longrightarrow$
 $Y \longrightarrow$
 $Z \longrightarrow$

Derivation

$A \longrightarrow XYZ$
 $a \longrightarrow A$
 $b \longrightarrow A$
 $c \longrightarrow A$

Parse Tree



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